University of Science and Technology at Zewail City

Deep Learning

CIE 555 - Spring 2026

Face Verification and Identification

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1 Introduction

Face verification is a critical task in biometric security and digital identity. Unlike Face Identification (which asks, "Who is this person?"), Face Verification asks a binary question: "Are these two images the same person?" The primary objective of this project is to develop a siamese network archtiecture for face verification and identification. The primary technical challenge in this project is the Open-Set Problem. Traditional convolutional neural networks (CNNs) are trained to classify images into a fixed number of categories. However, a robust face verification system must work for individuals it has never seen during training.

This requires the model to transition from Classification (memorizing names) to Metric Learning (learning the geometry of faces). The model must learn to extract a feature vector (embedding) such that different photos of the same person are mapped to nearby points in the vector space while photos of different people are mapped to distant points. This assignment explores two distinct loss functions:

- Model A: An original Convolutional Neural Network (CNN) built from first principles
- Model B: A transfer learning approach utilizing ResNet-18 pretrained backbone to leverage existing feature extractors for facial verification.

By implementing and comparing Contrastive and Triplet Margin Loss functions, this project aims to create a high-dimensional embedding space that generalizes to unseen identities, effectively separating Matches from Mismatches using an optimized Euclidean distance threshold.

2 Dataset and Preprocessing

2.1 Labeled Faces in the Wild (LFW)

The model was trained and evaluated using the **Labeled Faces in the Wild (LFW)** dataset. LFW is a widely recognized benchmark for face verification, containing over 13,000 images of faces collected from the web. The dataset is particularly challenging for metric learning due to its "unconstrained" nature, which includes:

- **Pose Variation:** Faces are captured from various angles, including frontal, profile, and three-quarter views.
- **Environmental Factors:** Significant differences in illumination, background clutter, and image resolution.
- **Identity Imbalance:** A high variance in the number of images per individual, necessitating a custom filtering strategy.

2.2 Data Filtering and Partitioning

To ensure the model could effectively learn the concept of identity through positive pairs, the raw data was filtered based on a frequency threshold:

- **Training/Validation Set:** Only individuals with at least 5 images were included. This ensures sufficient samples to generate multiple anchor-positive combinations.
- **Testing Set:** To evaluate the model's ability to generalize to unseen identities, individuals with fewer than 5 images were reserved for a separate testing partition. These identities were excluded entirely from the training phase.

Table 1 shows the total count of individuals as well as the total number of photos in each set. The individuals in each set were exclusive to that set meaning that no individual existed in both the training an validation sets.

However there was still one more issue which was how imbalanced the data was as demonstrated by Fig 1. To resolve this issue all individuals 5 or more images were chosen. For each individual 10 photos were selected. If the individual had more than 10 photos 10 were chosen at random. If the individual had less than 10 photos they were completed to 10 through augmentation.

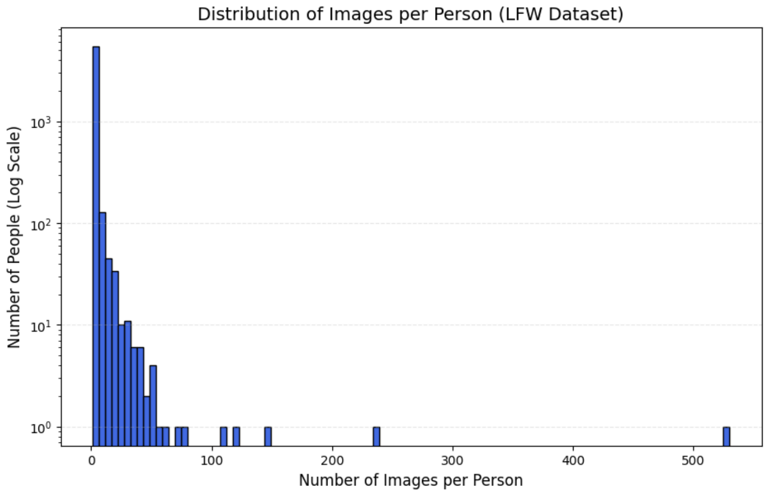


Figure 1: Frequency distribution of images per individual in the LFW dataset.

Table 1: Data Partitioning and Filtering Strategy

Dataset Split	**Identities**	**Images**	**Selection Logic**
Training	338	4835	≥ 5 images; used for gradient updates
Validation	85	1150	≥ 5 images; used for hyperparameter tuning
Testing (Zero-Shot)	5326	7248	< 5 images; completely unseen identities

Note: The testing set serves as a true measure of the model's ability to generalize facial features to individuals not present in the training manifold.

2.3 Preprocessing and Augmentation

Images were preprocessed to meet the input requirements of the ResNet18 backbone while improving model robustness:

- **Standardization:** All images were resized to 224×224 pixels and normalized using the ImageNet mean ($\mu = [0.485, 0.456, 0.406]$) and standard deviation ($\sigma = [0.229, 0.224, 0.225]$).
- **Augmentation:** Random horizontal flipping and color jittering were applied during training to simulate variations in orientation and lighting.

2.4 Siamese and Triplet Structure

Finally, the filtered data was restructured into specialized formats required by the loss functions. The contrastive loss function requires pairs of data with similar and dissimilar pairs. The triplet loss function requires three images one of them is an anchor, another is a positive sample (the same person as the anchor), the other is a negative sample (different person than the anchor).

3 Transfer Learning with ResNet18

The primary architecture used in this study is based on the **ResNet18** backbone. Utilizing a pre-trained model allows the system to benefit from a robust set of low-level and mid-level facial features already learned from the vast ImageNet dataset, significantly reducing training time and data requirements.

3.1 Architectural Adaptation

The ResNet18 architecture was adapted for Siamese and Triplet tasks by removing the standard $1,000$-class classification layer and appending a custom Embedding Head. This head consists of a fully connected layer that maps the global average-pooled features to a 128-dimensional manifold, followed by $L2$ normalization.

3.2 Two-Phase Training Strategy

To optimize performance and protect pre-trained weights from aggressive early gradients, a staged training approach was implemented:

1. **Phase 1 (Feature Extraction):** The ResNet18 backbone was frozen, keeping its weights static. Only the custom embedding head was trained using a relatively high learning rate (1×10^{-3}). This phase focuses on orienting the new head to the target dataset without distorting the pre-trained feature extraction capabilities.

2. **Phase 2 (Full Fine-Tuning):** The entire network, including the backbone, was unfrozen. A significantly lower learning rate (5×10^{-5}) was applied to carefully adjust the internal convolutional filters to the specific spatial nuances of the LFW faces.

3.3 Optimization and Regularization

The `Adam` optimizer was employed for both phases. To prevent overfitting during the fine-tuning stage, a weight decay of 1×10^{-4} was applied. Additionally, a `ReduceLROnPlateau` scheduler monitored the validation loss, decreasing the learning rate by a factor of 10 when the loss stagnated for two consecutive epochs. This ensures that the model converges to a stable local minimum in the embedding space.

4 Custom CNN Architecture

In addition to the ResNet18 transfer learning approach, a custom Convolutional Neural Network (CNN) was designed and trained from scratch. The primary objective of this experiment was to establish a baseline and evaluate the necessity of pre-trained deep architectures for facial verification on a relatively small-scale dataset like LFW.

4.1 Architectural Design

The `CNN_scratch` model follows a modular design consisting of a **Feature Extractor** and an **Embedding Head**:

- **Convolutional Layers:** The model utilizes three convolutional blocks with increasing filter sizes $(32, 64, 128)$. Each layer employs a 3×3 kernel with padding to maintain spatial resolution before pooling.

- **Normalization and Non-linearity:** *Batch Normalization* is applied after every convolution to stabilize learning and accelerate convergence. *ReLU* activation functions are used to introduce non-linearity.

- **Spatial Reduction:** *MaxPooling* layers reduce the spatial dimensions, while an *AdaptiveAvgPool2d* layer at the final stage compresses the spatial map into a 1×1 representation, regardless of input size.

- **Regularization:** A *Dropout* layer ($p = 0.1$) is integrated after the second block to mitigate overfitting, which is a significant risk when training a model from scratch on LFW.

4.2 Projection and Normalization

The final feature map is flattened and passed through a linear layer to project the features into a 128-dimensional embedding space. Similar to the ResNet18 implementation, an $L2$ normalization layer is applied at the output:

$$f(x) = \frac{x}{\|x\|_2} \tag{1}$$

This ensures that the loss functions (Contrastive and Triplet) operate on a unit hypersphere, making the Euclidean distance a consistent metric for similarity.

4.3 Training Rationale

Unlike the ResNet18 model, which benefited from ImageNet weights, the `CNN_scratch` model was initialized with random weights. To compensate for the lack of pre-trained knowledge, a higher learning rate (1×10^{-3}) was utilized alongside a `ReduceLROnPlateau` scheduler to dynamically adjust the step size as the validation loss plateaued.

5 Loss Function Comparison and Analysis

The success of a face verification system depends on the model's ability to learn a discriminative embedding space. This assignment evaluates two distinct approaches to metric learning: Contrastive Loss and Triplet Margin Loss.

5.1 Contrastive Loss: Pairwise Constraints

Contrastive loss treats verification as a pairwise distance optimization problem. For any two images, the model is penalized if the distance between positive pairs is large, or if the distance between negative pairs is smaller than a defined margin m.

The loss is defined as:

$$\mathcal{L}_{cont} = y \cdot d^2 + (1 - y) \cdot \max(0, m - d)^2 \tag{2}$$

While effective, Contrastive Loss imposes an absolute constraint: it attempts to force all positive pairs to a distance of zero. In high-dimensional facial data, this can be overly restrictive, as it does not account for natural intra-class variations such as lighting or aging.

5.2 Triplet Margin Loss: Relative Ranking

Triplet Loss shifts the focus from absolute distances to relative rankings. By processing a triplet (a, p, n), the model ensures that the anchor is closer to the positive sample than to the negative sample by at least a margin m:

$$\mathcal{L}_{trip} = \max(0, d(a, p)^2 - d(a, n)^2 + m) \tag{3}$$

This approach is generally superior for face verification because it allows the embedding manifold to be more flexible. The model does not need to collapse all images of a person into a single point; it only needs to ensure that the identity gap between the correct person and the other person is maintained.

5.3 The Role of the Margin (m)

In both functions, the hyperparameter m is crucial. It acts as a regularization term that prevents the model from becoming lazy. Without a margin, the model might find a trivial solution where it separates classes by a negligible distance. The margin forces the network to continue refining the weights until a robust, separable gap is formed in the hypersphere, directly improving the generalization accuracy on unseen identities in the testing set. In Contrastive Loss, the margin m acts as a hard boundary that defines the minimum absolute distance negative pairs must maintain, whereas in Triplet Loss, m represents a relative buffer ensuring that the anchor is closer to the positive sample than the negative sample by a specific gap.

5.4 Hyperparameter Selection: Margin

The hyperparameters $m = 1.0$ for Contrastive Loss and $m = 0.2$ for Triplet Loss were selected based on the geometric constraints of the L2-normalized embedding space.

- **Contrastive Margin (m):** Given that embeddings are mapped to a unit hypersphere, the maximum possible Euclidean distance is 2.0. A margin of 1.0 provides a robust push-force for dissimilar pairs without imposing an unreachable geometric constraint that would lead to gradient instability.
- **Triplet Margin (m):** In the triplet context, m represents the required relative gap between positive and negative distances. A value of 0.2 was chosen to ensure a clear identity separation while maintaining a feasible optimization landscape, preventing the vanishing gradient problem associated with overly aggressive margins.

6 Experimental Results and Analysis

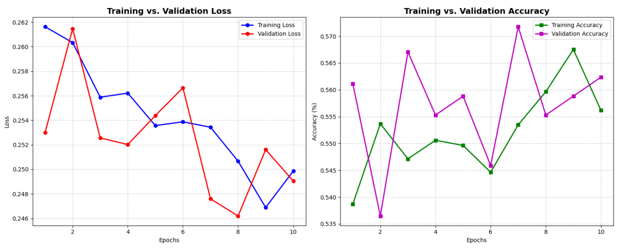
This section evaluates the performance of the four architectural configurations: `CNN-Scratch` (Contrastive/Triplet) and `ResNet18` (Contrastive/Triplet).

6.1 Training Dynamics and Convergence

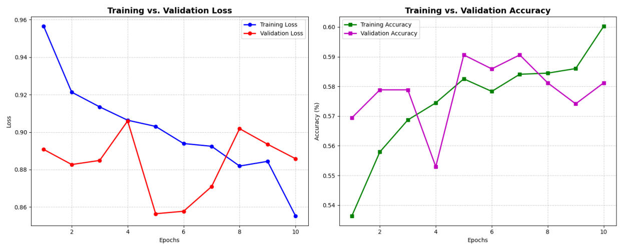
The training and validation curves provided significant insight into the learning capacity of each model.

- **CNN-Scratch:** The loss curves for the models trained from scratch exhibited higher variance and slower convergence as demonstrated by Fig. 2. Without the benefit of pre-trained spatial hierarchies, the model required more epochs to establish a stable embedding space.
- **ResNet18:** Both the Contrastive and Triplet variants of the ResNet18 model converged rapidly. The two-phase training approach (Phase 1: Head Warming, Phase 2: Full Fine-Tuning) proved highly effective, as evidenced by the sharp increase in accuracy once the backbone was unfrozen as demonstrated in Fig. 3.

The following figures provide a comparative analysis of the training dynamics for both the custom architecture and the transfer learning approach.

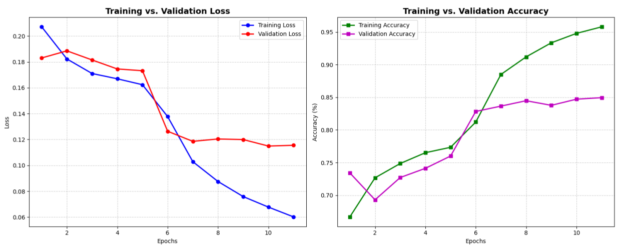


(a) CNN-Scratch: Contrastive Loss

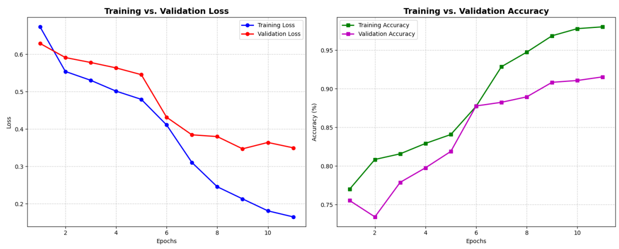


(b) CNN-Scratch: Triplet Loss

Figure 2: Training and Validation curves for the Custom CNN architecture trained from scratch.



(a) ResNet18: Contrastive Loss



(b) ResNet18: Triplet Loss

Figure 3: Training and Validation curves for the ResNet18 architecture using Transfer Learning.

Finally, we note how for both models the triplet loss reached better training and validation accuracy despite how they both started from nearly the same accuracy and were trained for the same number of epochs.

6.2 Embedding Space Analysis: Distance Distributions

The quality of the learned manifold is best visualized through the distribution of Euclidean distances for positive and negative pairs in the validation set.

- **Separation Gap:** The **ResNet18 Triplet** model demonstrated the most significant separation between classes as demonstrated by Fig 5. The histogram of positive pair distances was tightly clustered near 0.2, while negative pair distances were largely pushed beyond the 0.8 mark.
- **Overlap (Uncertainty):** The `CNN-Scratch` models showed a larger overlapping region in their histograms (Fig. 4), indicating a higher probability of False Accepts or False Rejects compared to the transfer learning approach.

The quality of the learned identity separation is visualized through the distribution of Euclidean distances for positive (same identity) and negative (different identity) pairs. A robust model is characterized by a significant gap or minimal overlap between these two distributions. To finalize the verification pipeline, an optimal decision threshold was empirically determined by analyzing the intersection of the positive and negative distance distributions. As seen in Figure 5, the ResNet18 Triplet model achieves its highest validation accuracy at a threshold of 0.88, effectively minimizing the Equal Error Rate where False Acceptance and False Rejection rates are balanced. This optimized boundary ensures that the system remains robust against identity spoofing while maintaining high sensitivity for legitimate matches.

(a) CNN-Scratch: Contrastive Histograms

(b) CNN-Scratch: Triplet Histograms

Figure 4: Euclidean distance distributions for the custom CNN architecture.

(a) ResNet18: Contrastive Histograms

(b) ResNet18: Triplet Histograms

Figure 5: Euclidean distance distributions for the ResNet18 architecture.

6.3 Comparative Performance Summary

Table 2 summarizes the peak performance identified for each model configuration.

Architecture	Loss Function	Max Val Accuracy
CNN-Scratch	Contrastive	56%
CNN-Scratch	Triplet	59%
ResNet18	Contrastive	85%
ResNet18	Triplet	91%

Table 2: Comparative metrics across all tested configurations.

6.4 Zero-Shot Evaluation on Unseen Identities

To test real-world applicability on unseen identities (Zero-Shot), two simulation scenarios were performed:

1. **Positive Match (Same Identity):** Two different images of a person not present in the training set were compared.
2. **Negative Match (Different Identities):** Images of two different individuals were compared.

The generalization capability of the models was evaluated using a Zero-Shot testing protocol. This test set consisted of identities with fewer than five images, ensuring the network had no prior exposure to these specific facial geometries during either the training or fine-tuning phases.

As illustrated in Figures 6 and 7, a comparison between the custom CNN and the ResNet18 (both trained with Triplet Loss) reveals a significant performance gap. The ResNet18 architecture demonstrates superior feature extraction: the Euclidean distance between similar images is notably tighter, while the distance between dissimilar images is more pronounced. This increased separation between the positive and negative distributions indicates that the ResNet18 model has learned a more robust, universal representation of facial features that generalizes effectively to strangers, whereas the custom CNN exhibits higher variance and narrower margins.

(a) CNN-Scratch: Zero-Shot

(b) ResNet18: Zero-Shot

Figure 6: Zero Shot Same Person Sample Test

(a) CNN-Scratch: Zero-Shot

(b) ResNet18: Zero-Shot

Figure 7: Zero Shot Different Person Sample Test

The resnet triplet model had a final test accuracy of 76% on the held out test set. The histogram in Fig. 8 shows the distance seperation histogram on the dataset. This indicates that the resnet triplet generalizes well to new data.

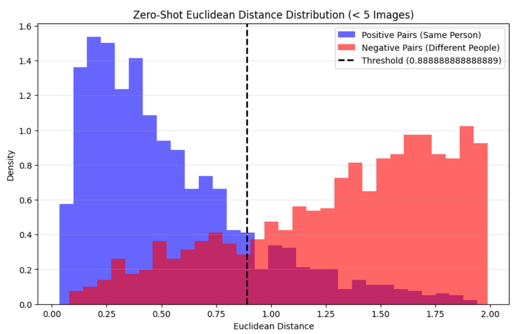


Figure 8: Zero-Shot Test Distribution Histogram for ResNet18 Triplet.

7 Conclusion

The results demonstrate that while custom architectures can learn facial representations, Transfer Learning with ResNet18 provides a superior baseline for biometric verification. Furthermore, Triplet Loss tends to create a more discriminative embedding space than Contrastive Loss by focusing on relative identity rankings, making it more robust to the natural variances found in the LFW dataset.